

# Finding Minimum Spanning Tree

## Definition:

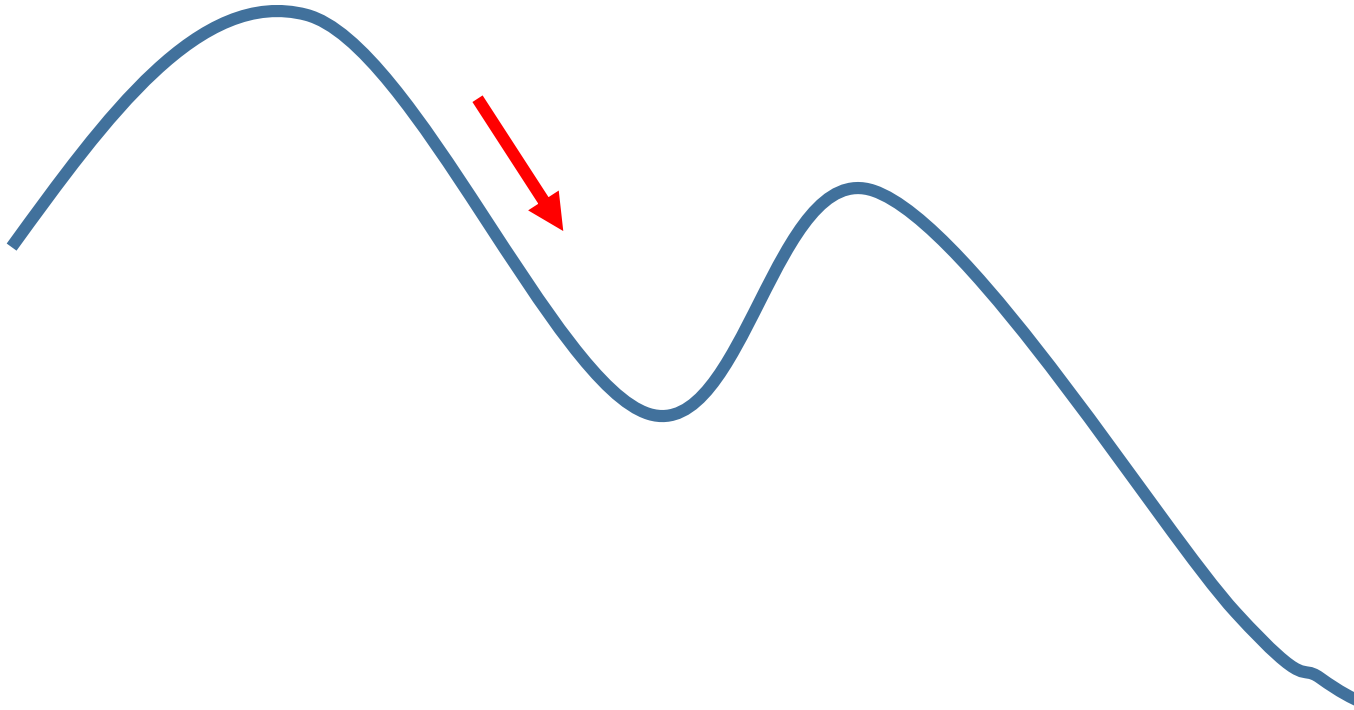
Let  $G = (V, E)$  be a graph, where  $E$  are the *possible* connections between vertices, and for each edge  $(u, v) \in E$  there is a weight  $w(u, v)$  which is the connection cost.

A *minimum spanning tree (MST)* is a subset  $T \subseteq E$  which connects all the vertices in  $V$ , such that the total weight

$$w(T) = \sum_{(u,v) \in T} w(u, v),$$

is the minimum.

# Greedy Choice Example – Finding Minimum



# Finding Minimum Spanning Tree

## A safe edge:

An edge  $e$  is a **safe edge** for a subset  $A$  of a minimum spanning tree if  $A \cup \{e\} \subseteq T$ , for some minimum spanning tree  $T$ .

# Finding Minimum Spanning Tree

## Definitions:

- 1) A *cut*  $(S, V - S)$  of a graph  $G = (V, E)$  is a partition of  $V$ .
- 2) An edge  $(u, v) \in E$  *crosses* a cut  $(S, V - S)$  if one of its endpoints is in  $S$  and the other in  $V - S$ .
- 3) Given a cut  $(S, V - S)$ , an edge  $(u, v)$  is *light* if its weight is minimal over the edges crossing the cut.
- 4) A cut  $(S, V - S)$  *respects* a set of edges  $A \subseteq E$ , if none of the edges in  $A$  crosses the cut.

# Finding Minimum Spanning Tree

## Theorem (safe edge criterion):

Let  $G = (V, E)$  be a connected graph, with a weight function:  
 $w: E \mapsto \mathbb{R}$ .

Let  $A \subseteq E$  be a subset of a minimum spanning tree  $T$  of  $G$ .

Let  $(S, V - S)$  be a cut of  $G$  that respects  $A$  and let  $(u, v)$  be a light edge crossing  $(S, V - S)$ .

Then,  $(u, v)$  is safe for  $A$ .

# Finding Minimum Spanning Tree

## Proof:

Let  $T$  be a MST of  $G$  such that  $A \subseteq T$ .

Let  $(S, V - S)$  be a cut of  $G$  that respects  $A$  and let  $(u, v)$  be a light edge crossing  $(S, V - S)$ . We will show that  $(u, v)$  is safe for  $A$ .

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If  $(u, v) \in T$  the proof is finished.

Otherwise  $((u, v) \notin T)$ , we will show that there exists another MST  $T'$ , such that  $A \cup \{(u, v)\} \subseteq T'$ .



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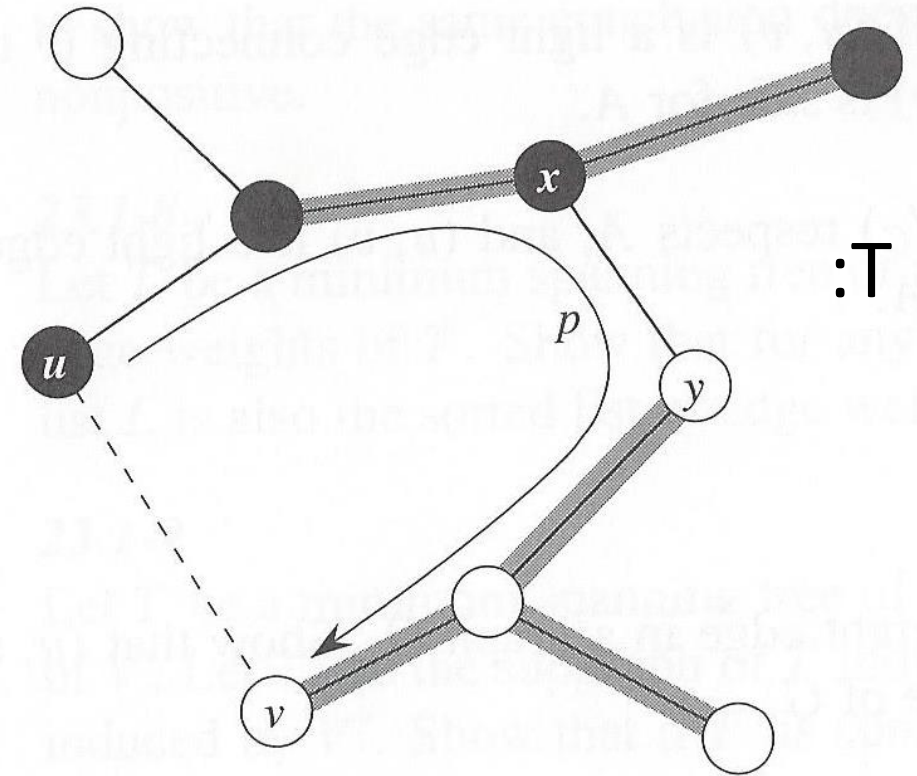
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Since  $T$  is a spanning tree there exists a path  $p$  in  $T$  that connects  $u$  and  $v$ .

# Finding Minimum Spanning Tree

Proof (cont.):

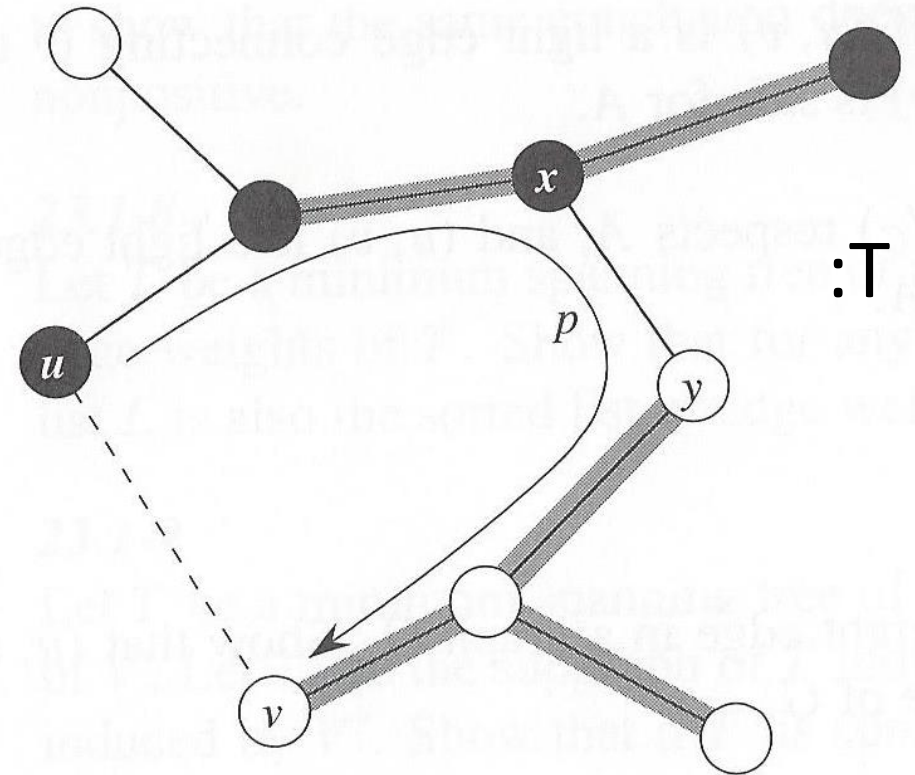
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# Finding Minimum Spanning Tree

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The edge  $(u,v)$  creates a cycle with the edges on the path  $p$ . Since  $u$  and  $v$  are on different sides of the cut  $(S, V - S)$ , there exists at least one edge on  $p$  that also crosses the cut. Let  $(x, y)$  be such edge.



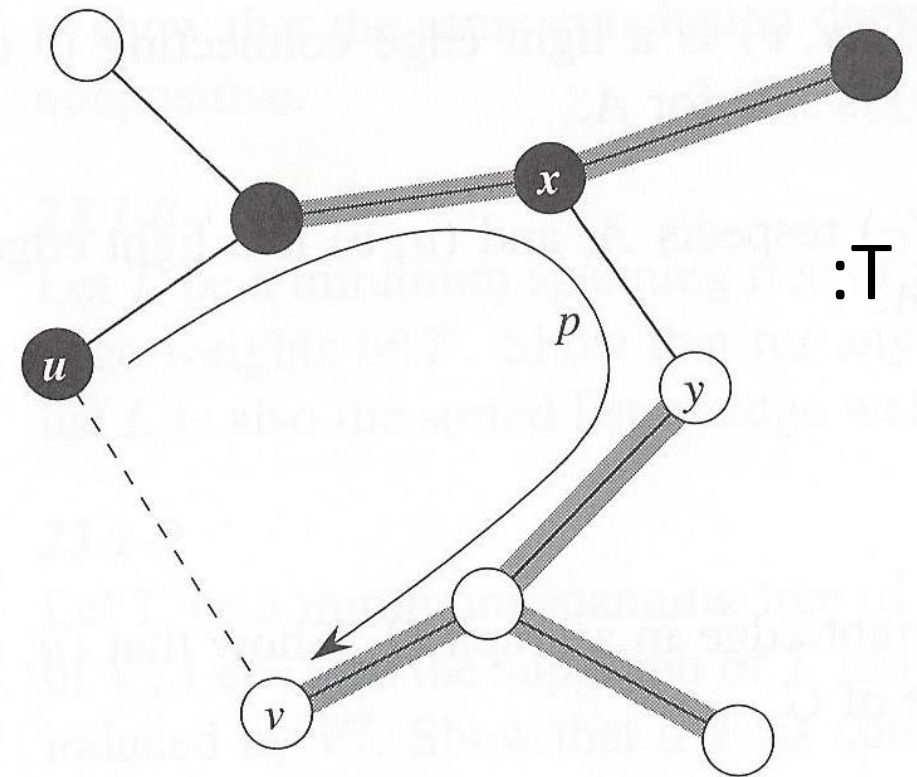
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Since  $u$  and  $v$  are on different sides of the cut  $(S, V - S)$ , there exists at least one edge on  $p$  that also crosses the cut. Let  $(x, y)$  be such edge.

Since  $(x, y)$  is on the only path in  $T$  from  $u$  to  $v$ , removing  $(x, y)$  decomposes  $T$  into 2 components. Adding  $(u, v)$  unites them to form a new spanning tree  $T'$ .



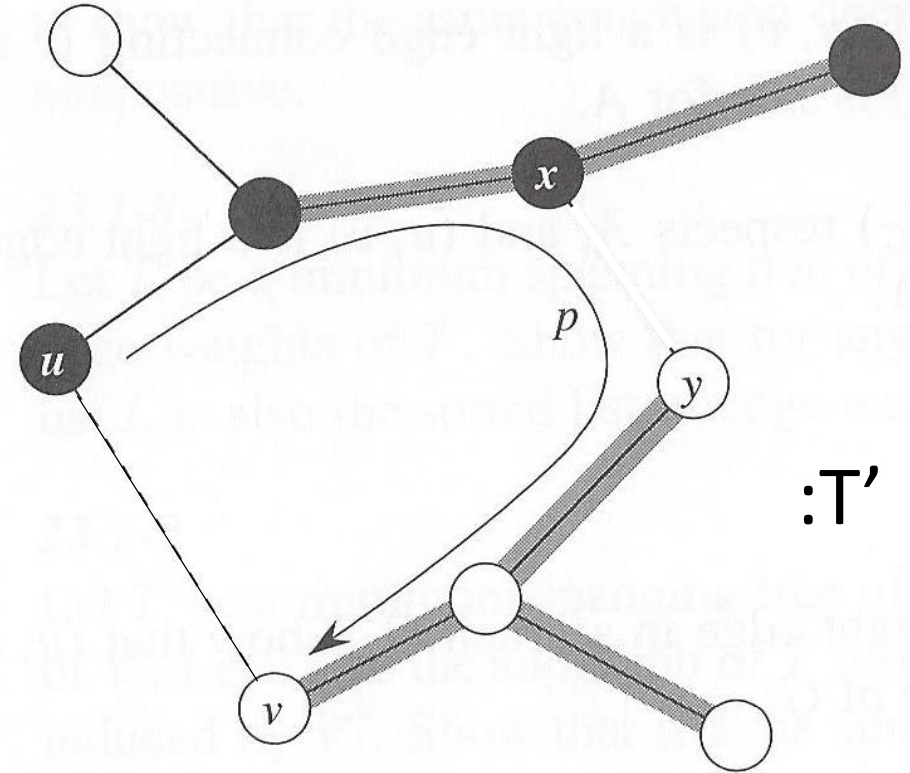
# Finding Minimum Spanning Tree

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We show that  $T'$  is an MST of  $G$ .

By the construction, we have:

$$T' = T \cup \{(u, v)\} - \{(x, y)\}.$$



# Finding Minimum Spanning Tree

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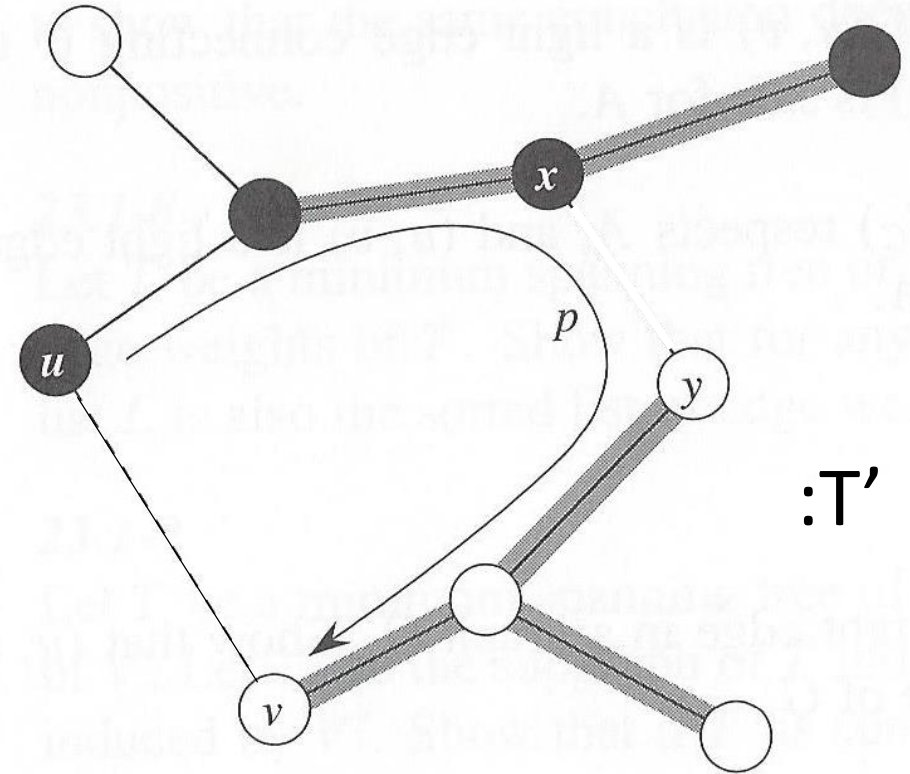
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$$w(T') = w(T) + w(u, v) - w(x, y)$$



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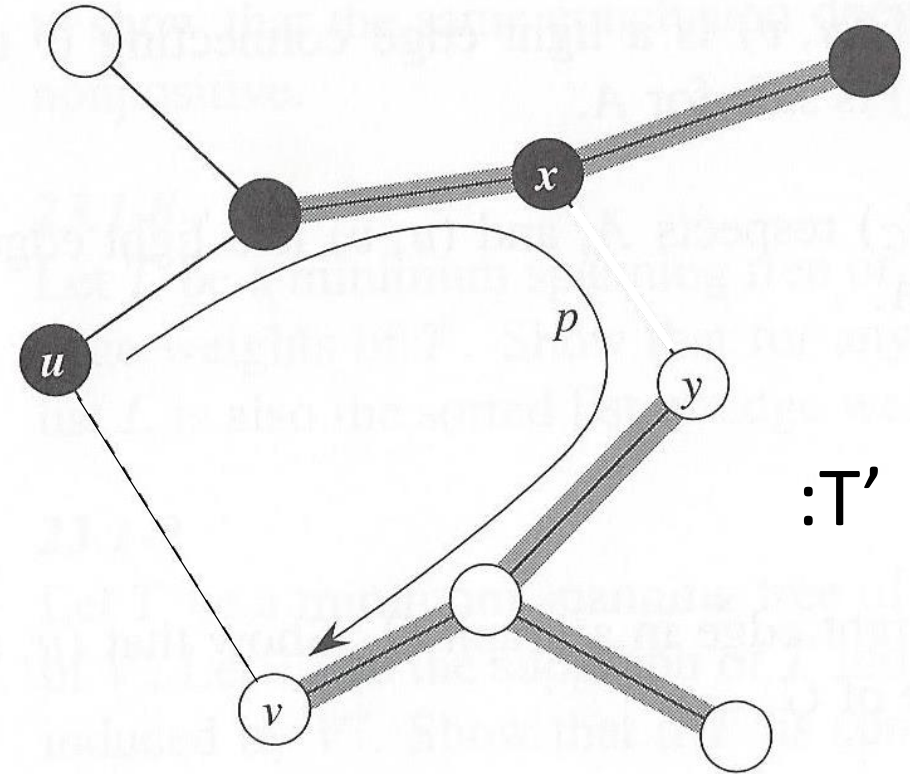
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But since  $w(u, v) \leq w(x, y)$ , we get:

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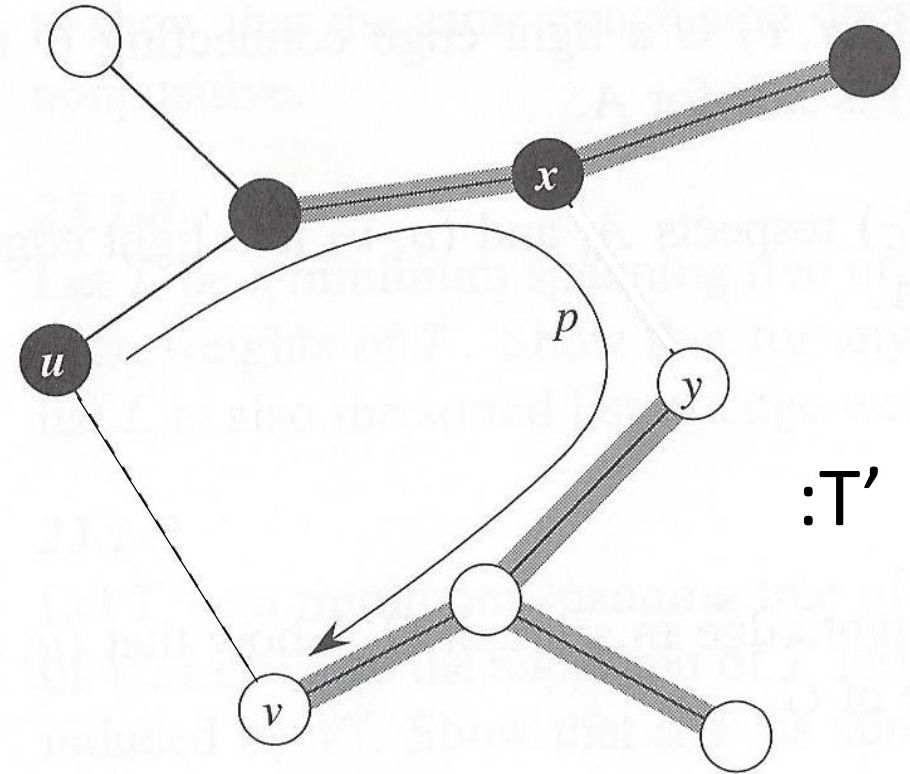
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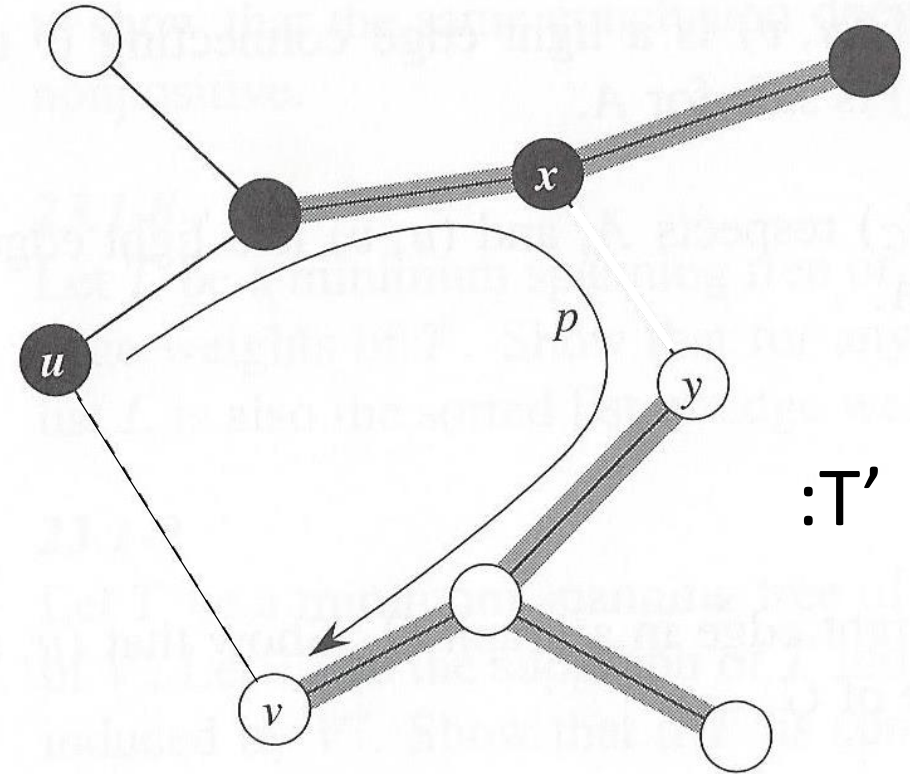
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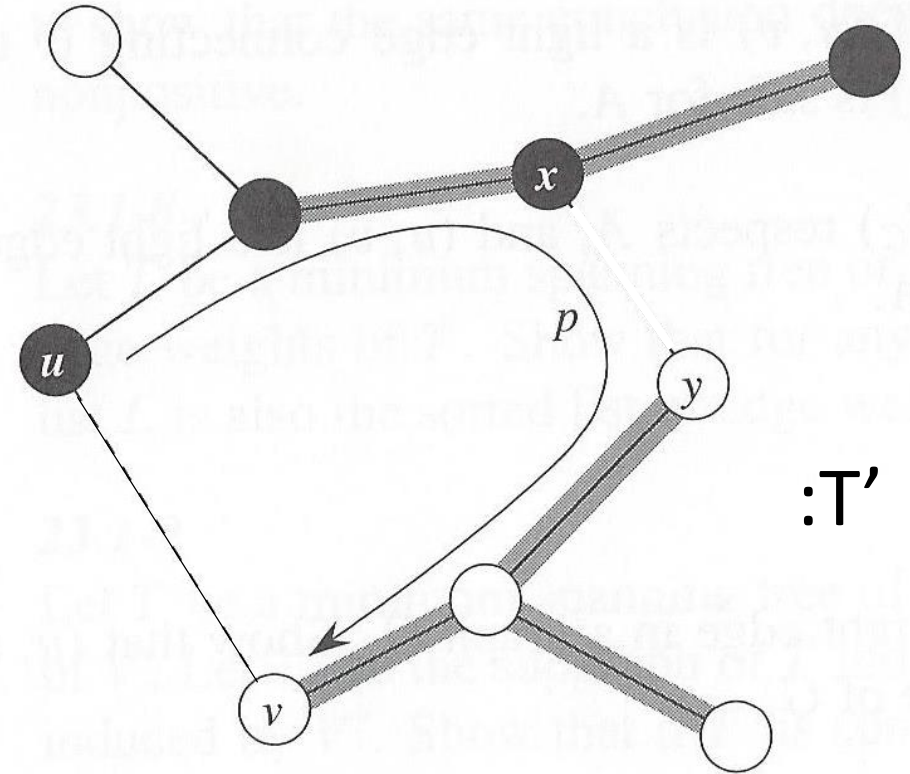
Therefore,  $w(T') = w(T)$  and  $T'$  is an MST.



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Proof (cont.):

We must still show that  $A \cup \{(u, v)\} \subseteq T'$ .



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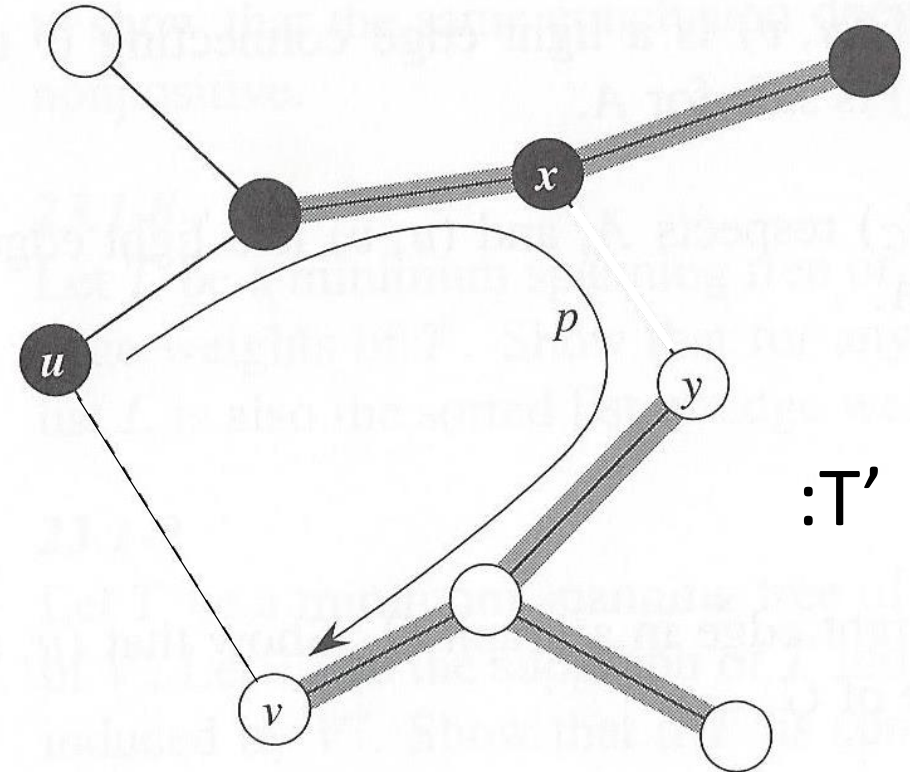
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$(u, v) \subseteq T'$  by the construction of  $T'$ .

$A \subseteq T'$  since every edge in  $T$  is also in  $T'$ , except for  $(x, y)$ , but  $(x, y) \notin A$ .

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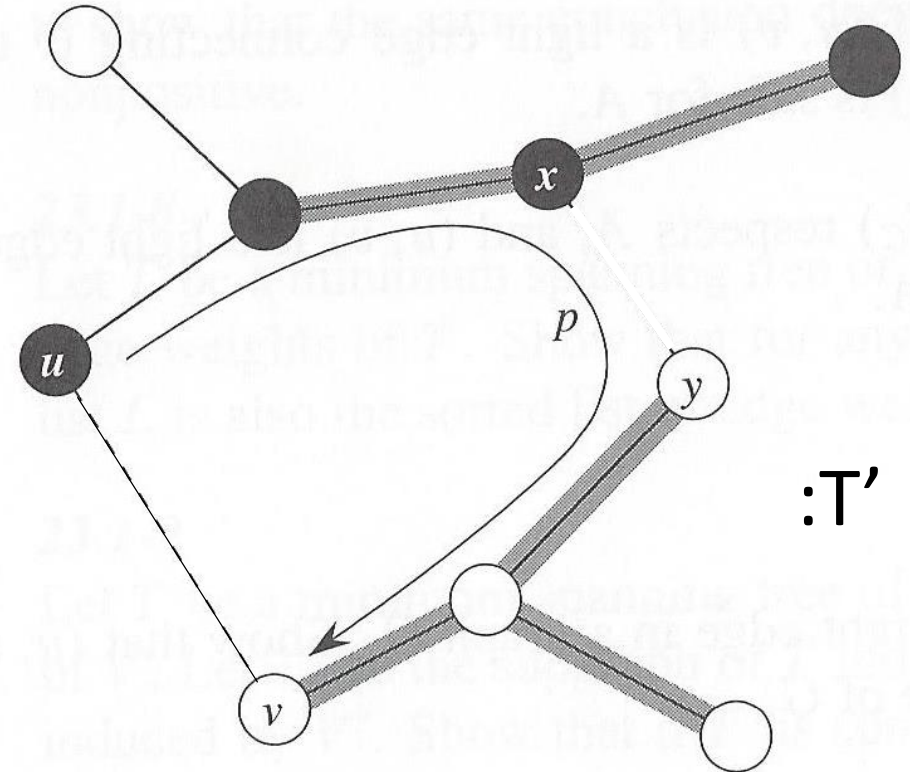
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Since the cut respects  $A$ , so no edge in  $A$  crosses the cut. However,  $(x, y)$  crosses the cut, therefore, cannot belong to  $A$ .



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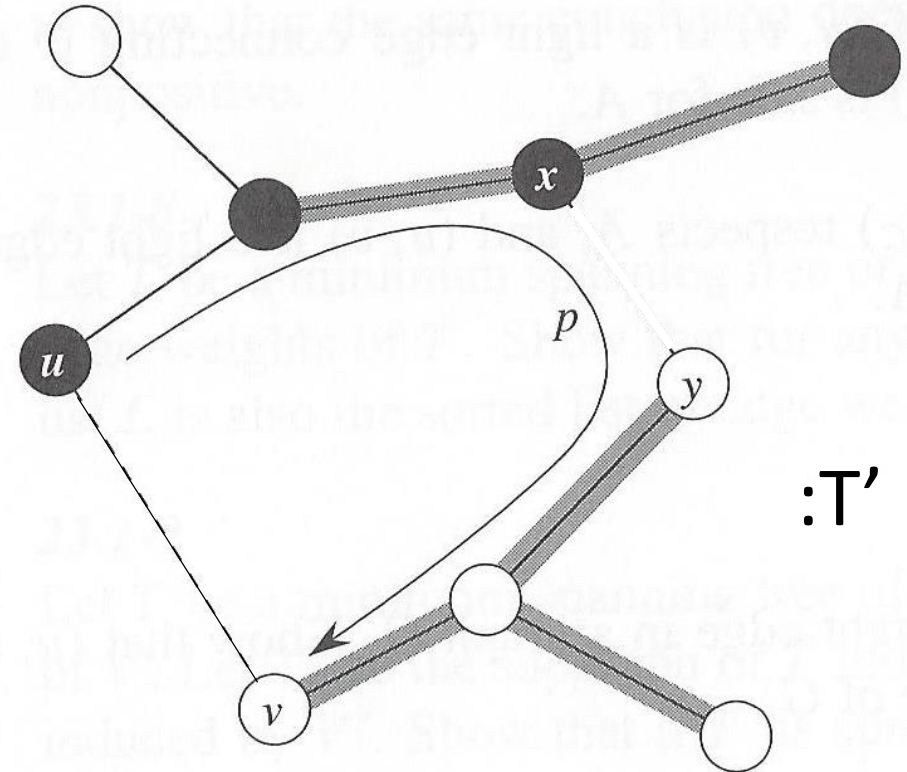
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Since the cut respects  $A$ , so no edge in  $A$  crosses the cut. However,  $(x, y)$  crosses the cut, therefore, cannot belong to  $A$ .

We have shown  $A \cup \{(u, v)\} \subseteq T'$ ,  $T'$  MST, so  $(u, v)$  is safe for  $A$ .



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## Prim's Algorithm Cost

| Command Number | Linear Queue    | Binary Heap           |
|----------------|-----------------|-----------------------|
| 1-3            | $\Theta( V )$   | $\Theta( V )$         |
| 4              | $\Theta(1)$     | $\Theta(\log V )$     |
| 5              | $\Theta( V )$   | $\Theta( V )$         |
| 6x{7}          | $\Theta( V ^2)$ | $\Theta( V  \log V )$ |
| 6x{8-11}       | $O( E )$        | $\Theta( E  \log V )$ |
| Total          | $\Theta( V ^2)$ | $\Theta( E  \log V )$ |